Software Requirements Specification
Long Island Traffic Accident Cluster Discovery System in R

AMS 597 Project Team

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1 Purpose

This project proposes an R-based analytical system that discovers meaningful patterns in Long Island traffic accidents using clustering techniques. The system is not a purely descriptive dashboard and it is not primarily a supervised prediction system. Its purpose is to discover latent accident patterns, accident scenario groups, and geographic hotspot structures along Long Island's highway corridors that are not obvious from simple count tables.

The project is restricted to Long Island in order to keep the scope geographically coherent, produce an original analysis grounded in the specific road network structure of Nassau and Suffolk counties, and make the final report easier to defend academically.

2 Problem Statement

The Long Island accident dataset contains mixed variable types:

- numeric variables such as temperature, visibility, wind speed, and pressure
- categorical variables such as weather condition and day versus night
- binary roadway context variables such as `Crossing`, `Junction`, and `Traffic_Signal`
- temporal information derivable from timestamps such as hour band, weekday, season, and rush-hour status
- a derived `Duration_Minutes` variable capturing traffic impact duration

Because of this structure, the main research problem is:

How can we discover interpretable and stable groups of Long Island accidents using clustering in R while respecting the mixed-type nature of the data and the corridor-dominated highway geography of Nassau and Suffolk counties?

3 Why Long Island Specifically

Restricting the work to Long Island is methodologically useful for four reasons:

- the county-specific scope reduces plagiarism risk because the project becomes more focused and less generic than a statewide or citywide analysis
- Long Island's road network is dominated by a small set of major east-west corridors (I-495, Southern State Parkway, Northern State Parkway, Sunrise Highway, Meadowbrook Parkway), which creates a structured geographic context well suited to spatial clustering
- Nassau and Suffolk county-level interpretation is meaningful in terms of land use, population density, and commuter patterns
- the dataset is large enough to support serious analysis while being small enough to avoid the extreme memory constraints of the full NYC region

The Long Island subset contains **33,375** accident records spanning from March 2016 to March 2023.

County	Rows
Nassau	19,390
Suffolk	13,838
Queens (western LI edge)	147
Total	**33,375**

Nassau and Suffolk are the primary analytical counties. The 147 Queens records at the western edge of Long Island are retained but are not the focus of county-level comparisons.

4 Dataset Profile Relevant to Clustering

4.1 Size and Severity Distribution

- Total rows: 33,375
- Date range: March 2016 to March 2023 (approximately 7 years)
- Severity 1: 161
- Severity 2: 28,357
- Severity 3: 3,938
- Severity 4: 919

Severity is heavily concentrated at level 2. This imbalance means severity should **not** be used as a clustering input variable if the goal is to interpret severity *after* the clusters are formed. Severity 4 represents a meaningful rare-risk category that post-cluster profiling can illuminate.

4.2 Dominant Road Corridors

The top accident streets are entirely highway corridors, which distinguishes Long Island from a city environment with dense intersections:

Road Segment	Accident Count
Southern State Pkwy Eastbound	3,859
Long Island Expressway Westbound	3,446
Southern State Pkwy Westbound	3,299
Long Island Expressway Eastbound	2,964
Northern State Pkwy Eastbound	1,559
Long Island Expressway (combined)	1,504
Northern State Pkwy Westbound	1,066
Sunrise Highway	804
Meadowbrook Pkwy Northbound	659
Meadowbrook Pkwy Southbound	640

This highway-dominated structure means spatial clustering will produce corridor-shaped hotspot zones rather than the radial or grid-like hotspots expected in an urban borough setting.

4.3 Top Accident Cities

City	Accident Count
Merrick	1,498
Roslyn Heights	1,297
Valley Stream	1,162
Huntington Station	1,144
Old Westbury	1,048
Wantagh	1,040
Bay Shore	1,029
Westbury	1,006
Melville	1,002
Jericho	1,002

4.4 Roadway Context Variable Prevalence

- `Junction = TRUE` for about 15.98% of rows
- `Crossing = TRUE` for about 10.11% of rows
- `Traffic_Signal = TRUE` for about 9.55% of rows
- `Stop = TRUE` for about 1.04% of rows
- `Station = TRUE` for about 0.70% of rows
- `Amenity = TRUE` for about 0.34% of rows

- `Railway = TRUE` for about 0.15% of rows

The lower prevalence of Railway and Amenity compared to a borough-dense urban setting reflects Long Island's primarily commuter-road infrastructure.

4.5 Temporal Patterns

Peak accident hours on Long Island follow the commuter pattern expected for a suburb-to-city corridor:

- morning rush peak at hours 07 and 08 (3,336 and 3,128 accidents)
- afternoon peak at hours 15 and 16 (2,402 and 2,363 accidents)
- secondary afternoon extension into hour 17 and 18

4.6 Duration Distribution

The `Duration_Minutes` variable (derived from start to end timestamp) has a median of approximately 78 minutes and a mean of approximately 449 minutes. The mean is inflated by extreme outliers (some records exceeding 1,000,000 minutes, which reflect data entry anomalies). Duration should either be log-transformed or capped before use as a clustering feature.

4.7 Missingness

- `End_Lat` / `End_Lng`: about 39.2% missing (these are not used as clustering inputs)
- `Precipitation(in)`: about 20.1% missing
- `Wind_Chill(F)`: about 15.0% missing
- `Weather_Condition`: about 0.02% missing (fewer than 10 rows)
- `Sunrise_Sunset` and twilight fields: about 0.02% missing

Implications:

- `Precipitation(in)` should not be a core clustering feature unless there is a clear imputation strategy; 20% missingness is too high for routine median imputation
- `Wind_Chill(F)` is largely redundant with `Temperature(F)` and has 15% missingness; exclude from the core model
- `End_Lat` and `End_Lng` are excluded from all clustering as they are missing for 39% of rows and are not needed when `Start_Lat` and `Start_Lng` are used for spatial clustering

5 Research-Backed Method Selection

5.1 Why Plain k-Means Should Not Be the Main Method

The standard R implementation of `kmeans()` expects numeric input. The Long Island accident data is not naturally numeric-only. It includes factors, binaries, time-derived categories, and mixed roadway conditions.

Therefore:

- plain k-means is not the best primary method for the full research workflow
- forcing all variables into crude numeric codes would damage interpretability
- k-means can still be used later as a reduced numeric-only comparison, but not as the main solution

5.2 Why Full Gower Plus Hierarchical Clustering or PAM Is Not the Main Method

The function `cluster::daisy()` supports mixed variable types through Gower-style dissimilarities. For the Long Island dataset of 33,375 rows:

- pairwise distances = approximately 556,546,875
- storing the distance matrix as doubles would require approximately 4.15 GiB

While this is more tractable than the NYC scale, 4.15 GiB is still prohibitive for most computing environments without dedicated memory allocation. Therefore:

- full `daisy()` plus `agnes()` or `pam()` on all 33,375 rows is not recommended as the primary workflow
- this strategy may be used on a smaller sample for sensitivity analysis or comparison

5.3 Why CLARA Is Not the Best Primary Method

`cluster::clara()` is designed for large applications. However, its input expects numeric or logical data. For this project, the mixed categorical structure would need extra encoding before clustering.

Therefore:

- CLARA is a reasonable fallback for a numeric or dummy-encoded workflow
- it is not the cleanest primary method for mixed accident records

5.4 Recommended Primary Method: k-Prototypes

The recommended primary method is `clustMixType::kproto()`.

Reasons:

- it is designed for mixed-type data
- it accepts numeric and factor variables together
- it supports `type = "gower"`, which range-normalizes numeric distances
- the package ecosystem supports validation and cluster stability analysis
- the associated R Journal paper explicitly argues that all-pairs mixed-distance approaches are not feasible for large datasets, whereas k-prototypes scales linearly in the number of observations

5.5 Recommended Secondary Method: DBSCAN

The recommended secondary method is `dbscan::dbscan()` for spatial hotspot discovery along Long Island's highway corridors.

Reasons:

- Long Island accident hotspots are expected to follow elongated east-west highway shapes (LIE, Southern State, Northern State) rather than compact urban grid clusters
- DBSCAN is designed for dense regions with arbitrary shape, making it appropriate for corridor-style hotspots
- DBSCAN naturally labels sparse non-corridor points as noise
- the highway-dominated setting means a small number of meaningful clusters is expected, which aligns with DBSCAN's behavior

5.6 Final Method Decision

The best overall project design is a two-layer clustering system:

1. **Scenario clustering**: use `kproto(type = "gower")` on mixed contextual variables
2. **Spatial clustering**: use `dbscan()` on projected start coordinates, with emphasis on highway corridor shape discovery

This is better than trying to force a single algorithm to answer every question.

6 Recommended Research Framing

The strongest topic statement is:

Cluster-Based Discovery of Accident Scenarios and Spatial Hotspots on Long Island Using R

Recommended research questions:

- What latent accident scenario clusters exist on Long Island when accidents are grouped by time, weather, and roadway context?
- How do these clusters differ between Nassau and Suffolk counties?
- Which discovered clusters contain a higher share of severe accidents (Severity 3 or 4)?
- Where are the densest accident hotspot corridors on Long Island, and do they align with known high-volume highways such as the LIE and Southern State Parkway?
- Do the spatial hotspot zones correspond to particular scenario clusters, for example rush-hour highway scenarios versus off-peak surface-road scenarios?

7 What the Project Should and Should Not Do

7.1 What We Should Do

- use `li_accidents_clean.csv` as the official project dataset
- build one mixed-type scenario clustering pipeline in R
- build one separate spatial hotspot clustering pipeline in R focused on LI highway corridor shapes
- exclude severity from the clustering input matrix and use it only afterward for interpretation
- engineer interpretable time features from `Start_Time`
- collapse rare weather categories into an `Other` group
- use county (Nassau vs Suffolk) mainly for reporting and comparison
- validate cluster quality and stability rather than choosing cluster counts arbitrarily
- document the `Duration_Minutes` outlier situation and decide on a treatment strategy

7.2 What We Should Avoid

- do not run plain k-means on the raw mixed dataset
- do not use `ID`, `Description`, `Street`, or raw text as clustering inputs

- do not include `Severity` as a clustering feature if the goal is to interpret severity after clustering
- do not build a full all-pairs Gower distance matrix for all 33,375 rows without memory planning
- do not overclaim causality
- do not rank counties as more dangerous without traffic exposure denominators
- do not impute `Precipitation(in)` without justification given the 20% missingness rate

8 System Scope

8.1 In Scope

- ingestion of `li_accidents_clean.csv`
- cleaning and feature engineering in R
- mixed-type clustering for accident scenario discovery
- density-based spatial clustering for highway hotspot discovery
- cluster validation and stability assessment
- Nassau versus Suffolk county comparison of discovered clusters
- severity profiling after clustering
- exports, tables, and charts for the final report

8.2 Out of Scope

- causal inference
- real-time prediction
- live traffic feed integration
- deep learning on text descriptions
- national or statewide generalization
- NYC borough comparisons

9 Stakeholders

The main stakeholders are:

- the student research team
- the course instructor
- transportation and safety audiences who may read the final report
- the class presentation audience

10 Functional Requirements

FR-1 Data ingestion. The system shall load `li_accidents_clean.csv` in R and preserve the original columns before preprocessing.

FR-2 Data cleaning. The system shall parse timestamps, handle missing values, and convert binary and categorical variables to suitable types.

FR-3 Time feature engineering. The system shall derive hour, hour band, weekday, weekend flag, month, season, and rush-hour status from `Start_Time`.

FR-4 Duration treatment. The system shall document the `Duration_Minutes` outlier distribution and apply either a log transform or a cap before any use in clustering.

FR-5 Weather grouping. The system shall collapse rare weather categories into a smaller interpretable set.

FR-6 Scenario matrix creation. The system shall build a mixed-type feature matrix for scenario clustering.

FR-7 Leakage control. The system shall exclude `Severity`, IDs, free-text fields, and raw coordinates from the scenario clustering matrix.

FR-8 Scenario clustering. The system shall run `kproto()` with repeated starts to reduce unstable initializations.

FR-9 Cluster-count selection. The system shall evaluate candidate cluster counts, recommended as $k = 3$ through 8, using validation output.

FR-10 Stability analysis. The system shall perform bootstrap-based cluster stability checks.

FR-11 Cluster profiling. The system shall summarize each discovered cluster using dominant conditions, numeric summaries, county composition (Nassau vs Suffolk), and severity composition.

FR-12 Spatial hotspot clustering. The system shall run DBSCAN on projected start coordinates and identify highway corridor hotspot clusters and noise points.

FR-13 Export. The system shall export cluster-labeled outputs and summary tables for reporting.

11 Non-Functional Requirements

NFR-1 Reproducibility. The workflow shall be reproducible through R scripts or R Markdown with fixed random seeds where needed.

NFR-2 Interpretability. The clustering approach shall favor explainability over novelty.

NFR-3 Scalability. The main workflow shall operate on the full Long Island subset without requiring a full pairwise distance matrix.

NFR-4 Maintainability. The code shall be split into readable preprocessing, clustering, validation, and export stages.

NFR-5 Academic defensibility. The algorithm choice shall be justified with source-backed reasons and clear methodological arguments.

NFR-6 Ethical reporting. Results shall be presented as exploratory pattern discovery, not proof of causation.

12 Data Requirements

12.1 Recommended Inputs for Scenario Clustering

Recommended features:

- `hour_band`
- `weekday`
- `weekend`
- `season`
- `rush_hour`
- `Sunrise_Sunset`
- `weather_group`
- `Temperature(F)`
- `Humidity(%)`
- `Pressure(in)`
- `Visibility(mi)`
- `Wind_Speed(mph)`
- `Crossing`
- `Junction`

- `Traffic_Signal`
- `Stop`
- `Railway`
- `Amenity`
- `Station`

12.2 Variables to Treat Carefully

- `Precipitation(in)`: 20.1% missing, so exclude from the core model unless a justified imputation strategy is applied
- `Wind_Chill(F)`: 15.0% missing and largely redundant with `Temperature(F)`; exclude from the core model
- `Duration_Minutes`: heavy outliers present (max exceeds 2 million minutes); apply log transform or cap before use
- `Distance(mi)`: may reflect reporting behavior or event extent more than accident scenario type
- `County`: useful for post-cluster comparison between Nassau and Suffolk, but not ideal as a dominant clustering input

12.3 Variables to Exclude From Scenario Clustering

- `Severity`
- `ID`
- `Source`
- `Description`
- `Street`
- `City`
- `Zipcode`
- `nzip`
- `Start_Lat`
- `Start_Lng`
- `End_Lat`
- `End_Lng`

13 Detailed Analytical Design

13.1 Scenario Clustering

Goal:

- discover accident archetypes such as rush-hour highway clusters, night low-visibility clusters, and junction-heavy surface-road clusters that are specific to Long Island's commuter-corridor geography

Recommended method settings:

- algorithm: `kproto()`
- distance type: `type = "gower"`
- candidate cluster counts: $k = 3 : 8$
- repeated starts: at least `nstart = 5`, preferably `10`
- missing-data handling: internal imputation or a clearly documented preprocessing step

13.2 Spatial Hotspot Clustering

Goal:

- discover dense accident corridors along Long Island's major east-west highways (I-495 LIE, Southern State Pkwy, Northern State Pkwy, Sunrise Hwy) and north-south connectors (Meadowbrook Pkwy, Wantagh Pkwy, Bethpage State Pkwy)

Recommended method settings:

- algorithm: `dbscan()`
- inputs: projected start coordinates in approximate kilometer units
- parameter tuning: choose `eps` from a k-nearest-neighbor distance plot instead of guessing blindly; Long Island's elongated highway corridors may require a slightly larger `eps` than a dense urban setting
- inspect the percentage of noise points and the number and shape of clusters

13.3 Interpretation Strategy

After clustering:

- join cluster labels back to the cleaned dataset
- compute severity distribution by cluster
- compute county distribution (Nassau vs Suffolk) by cluster
- compare day versus night and weather composition by cluster
- compare hotspot corridor assignments with scenario clusters where useful
- assign plain-English names to the discovered clusters (for example: *AM Rush LIE Corridor*, *Weekend Surface Road*, *Night Low-Visibility*, *PM Merge Zone*)

14 Validation Strategy

The project should not stop at obtaining cluster labels. It must show why the clusters are acceptable.

14.1 Validation for k-Prototypes

Use:

- `validation_kproto()` for internal validation across candidate values of k
- `stability_kproto()` for bootstrap stability checks
- `summary()` and `clprofiles()` for interpretation

Decision rule:

- choose a solution that balances interpretability, validation quality, and stability
- avoid cluster solutions dominated by tiny clusters unless they represent a meaningful rare-risk pattern (such as a Severity 4 cluster)

14.2 Validation for DBSCAN

Use:

- kNN distance plots for parameter selection
- hotspot size inspection
- map-based inspection of cluster shapes, particularly whether clusters follow highway alignments
- percentage of noise points

15 Use Cases

UC-1 Full workflow execution. A researcher runs the full preprocessing, scenario clustering, hotspot clustering, and export pipeline on `li_accidents_clean.csv`.

UC-2 County comparison. A researcher compares how Nassau and Suffolk counties are distributed across the discovered scenario clusters.

UC-3 Severity interpretation. A researcher evaluates which discovered clusters contain a larger proportion of Severity 3 and Severity 4 accidents.

UC-4 Hotspot analysis. A researcher maps DBSCAN hotspot assignments and interprets dense highway accident zones along the LIE and Southern State Pkwy.

16 Acceptance Criteria

The project is successful if it can:

- load and preprocess `li_accidents_clean.csv` reproducibly in R
- generate an interpretable mixed-type clustering solution for Long Island accident scenarios
- justify the selected cluster count with validation output
- demonstrate cluster stability analysis
- produce a DBSCAN hotspot analysis showing corridor-aligned clusters
- connect cluster results back to severity and county patterns
- generate report-ready tables and figures

17 Risks and Mitigations

Risk 1: Mixed-type distance is handled poorly.
Mitigation: use `kproto()` rather than forcing the data into plain k-means.

Risk 2: Highway corridors dominate the scenario clusters.
Mitigation: keep spatial clustering separate from scenario clustering; do not include street names or raw coordinates in the scenario feature matrix.

Risk 3: Severity imbalance distorts the clusters.
Mitigation: do not cluster on severity directly; profile it afterward.

Risk 4: Rare weather labels destabilize the solution.
Mitigation: collapse infrequent weather conditions into an `Other` category.

Risk 5: High precipitation missingness weakens the model.
Mitigation: exclude precipitation from the core design (20.1% missing) or justify imputation clearly.

Risk 6: Duration outliers distort the numeric distance.
Mitigation: log-transform or cap `Duration_Minutes` before including it in the feature matrix.

Risk 7: The algorithm choice is challenged in review.
Mitigation: justify the method choice with official R documentation and package literature.

18 Phase-by-Phase Implementation Breakdown

This section translates the SRS into an execution plan that can be implemented step by step.

18.1 Phase 1: Repository Setup and Data Organization

Objective: prepare a clean project structure for Topic 3 and isolate the Long Island clustering work from other project experiments.

Tasks:

- keep the Long Island CSV (`li_accidents_clean.csv`), the SRS source, the compiled PDF, and the R clustering script in the topic-specific folder
- keep the project on the dedicated `topic-3-nyc-accident-clustering` branch
- remove temporary helper scripts and LaTeX build artifacts from the working directory

Deliverables:

- a clean Topic 3 folder
- a clean git status before commit

Done criteria:

- only the files needed for Topic 3 remain as intended deliverables
- the branch is ready for commit and push

18.2 Phase 2: Data Preparation

Objective: build an analysis-ready Long Island accident dataset in R.

Tasks:

- load `li_accidents_clean.csv`
- parse `Start_Time` and related time fields
- derive hour, hour band, weekday, weekend, month, season, and rush-hour indicators

- address `Duration_Minutes` outliers (log transform or cap)
- convert roadway flags and category variables into factors
- inspect missingness and decide which variables will be excluded, retained, or imputed
- collapse rare weather conditions into a smaller grouped factor

Deliverables:

- cleaned analysis-ready R data frame
- a documented feature list for clustering
- a short data-cleaning summary for the report

Done criteria:

- the dataset can be loaded and transformed reproducibly
- the final clustering input variables are fixed and justified

18.3 Phase 3: Scenario Clustering with k-Prototypes

Objective: discover mixed-type accident scenario groups across Long Island.

Tasks:

- build the final mixed-type feature matrix
- exclude `Severity`, free-text columns, identifiers, and raw coordinates from the scenario matrix
- run candidate `kproto()` solutions for several values of k
- compare solutions using internal validation
- run repeated starts so the chosen clustering is not dependent on one weak initialization

Deliverables:

- candidate scenario clustering results
- validation output for the tested values of k
- a selected final scenario clustering model

Done criteria:

- one final scenario clustering solution is selected and justified
- the chosen clusters are interpretable and not trivially driven by one variable

18.4 Phase 4: Cluster Validation and Stability

Objective: prove that the selected scenario clusters are academically defensible.

Tasks:

- apply `validation_kproto()` to compare candidate solutions
- apply `stability_kproto()` using bootstrap-based resampling
- inspect cluster sizes to ensure the solution is not dominated by very small clusters
- profile the clusters to confirm that each group has a meaningful interpretation tied to Long Island's highway commuter context

Deliverables:

- validation results for the selected cluster count
- stability metrics
- cluster profile summaries

Done criteria:

- the final cluster choice is supported by validation and stability evidence
- each cluster can be described in plain language relevant to Long Island travel patterns

18.5 Phase 5: Spatial Hotspot Clustering with DBSCAN

Objective: identify dense geographic accident hotspot zones along Long Island's highway corridors.

Tasks:

- convert start coordinates into projected distance units suitable for spatial clustering
- inspect k-nearest-neighbor distance behavior and choose `eps` from the elbow of the kNN distance plot
- tune `minPts` based on the expected corridor density
- run DBSCAN on the projected coordinates
- separate hotspot clusters from noise points and verify that corridor shapes (LIE, Southern State, Northern State) are detected

Deliverables:

- final DBSCAN hotspot assignments

- hotspot counts and county summaries
- parameter-selection justification

Done criteria:

- hotspot clusters are meaningful and align with known high-volume LI roads
- the level of noise is acceptable and explainable

18.6 Phase 6: Post-Clustering Interpretation

Objective: connect the discovered clusters back to real Long Island accident patterns.

Tasks:

- join scenario cluster labels back to the cleaned dataset
- compare severity distribution by scenario cluster
- compare Nassau versus Suffolk composition by scenario cluster
- compare day versus night and weather composition by cluster
- compare hotspot clusters with scenario clusters where useful
- assign plain-English names to the major cluster types (for example: *AM Rush Highway*, *PM Merge Zone*, *Night Low-Visibility Surface Road*, *Weekend Off-Peak*)

Deliverables:

- cluster interpretation tables
- severity-by-cluster summaries
- Nassau/Suffolk-by-cluster summaries
- named cluster narratives for the report

Done criteria:

- the final report can explain what each cluster represents in terms of Long Island commuter and road patterns
- the interpretation is tied back to transportation and safety context

18.7 Phase 7: Reporting, Branching, and Submission

Objective: package the work cleanly for review and submission.

Tasks:

- finalize the SRS in LaTeX
- compile the final PDF
- keep the R script aligned with the documented workflow
- commit the Topic 3 files on the dedicated branch
- push the branch to the remote repository

Deliverables:

- final `.tex` SRS
- compiled PDF
- R clustering workflow
- pushed git branch for Topic 3

Done criteria:

- the branch contains only the intended Topic 3 deliverables
- the remote branch is available for sharing, review, or merge

19 Final Recommendation

If the project needs one strong and defensible direction, the best choice is:

Build a Long Island accident clustering system in R with two complementary components:

- **k-prototypes** for mixed-type accident scenario discovery across Nassau and Suffolk counties
- **DBSCAN** for spatial hotspot discovery along Long Island's major highway corridors

This direction is stronger than a simple descriptive analysis, more appropriate than plain k-means for the available variables, and more geographically grounded than a generic statewide or NYC-wide approach. The corridor-dominated geography of Long Island makes DBSCAN's ability to detect elongated dense clusters a natural methodological fit.

20 References

References

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